RAUL ESTEBAN MINA RIASCOS

Process & Systems Engineering Services – P&ID, equipment design, simulation, site integration

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Chemical engineer (MSc) with six years in oil & gas services, energy technology and climate technology. I work as an engineering service partner: I take a scope from the process basis to the drawings, the calculations and the site, and I hand it over documented. This page lists what I deliver; page 2 lists the projects where I delivered it.

SERVICE LINES

1. Process & system engineering

- P&ID and PFD packages, design basis, line and equipment lists
- System definition; integration into existing plants (as-built, reverse engineering)
- Equipment and instrumentation selection, cause-and-effect logic

2. Thermal & mechanical design

- Separators, gun barrels, treaters, KO drums, storage and flowback systems
- Shell & tube exchangers (TEMA), flare systems
- Pressure equipment under ASME U, R, S stamps; API and ISO compliance

3. Simulation & flow assurance

- HYSYS, PIPESIM, OLGA, HTRI, ANSYS, MATLAB
- Hydrate formation, erosion and heat-loss prediction, multiphase flow, relief sizing
- Flare radiation, noise studies; blending and dilution simulations

4. 3D, CAD & documentation

- AutoCAD Plant 3D and Inventor: skids, pipe racks, anchor blocks, layouts
- Welding procedures, pressure test procedures, NACE material selection
- Standardised drawing sets and calculation reports across projects

5. Project & interface management

- 35+ well testing projects co-managed for international operators
- Procurement support, vendor documentation, construction supervision
- API and ISO 9001/14001 audits with zero findings

6. Digital & AI automation

- Python tooling for rigorous mechanical and process calculations
- Documentation and data pipelines (knowledge graphs, RAG)
- AI-assisted engineering documentation with human review

MAPPING TO LEAD PROCESS / SYSTEM AND INSTRUMENTATION DESIGN ENGINEER (R169394)

System definition and P&ID for upgrades	Service 1 – 50+ P&ID/PFD packages released at Weatherford
Auxiliary systems: lube oil, seal gas, fuel gas, cooling water	Service 2 – thermal and mechanical design of auxiliary and utility circuits around surface equipment
Instrumentation selection and integration	Services 1 and 4 – selection, layouts and as-built integration
Simulation (Hysys, Matlab/Simulink, FlowMaster)	Service 3 – HYSYS, PIPESIM, OLGA, HTRI, MATLAB
Site surveys, reverse engineering, installation support	Services 1 and 5 – field integration and construction supervision
Climate Technology Solutions	DAC process development and scale-up at Neocarbon (2025)

WORKING MODEL

Fixed-scope deliverable, hourly support, or embedded in your engineering team for the duration of a project. Based in Berlin, mobile across the EU for site work. Documentation in English, Spanish or German.

Selected project experience

What the services on page 1 produced, by client

INTERCOL S.A. — DESIGN AND CALCULATION

- Three-phase separators 60″ x 15′ ANSI 900, skid-mounted and trailerised (for Weatherford)
- Shell & tube heat exchangers 64″ x 24′, 4128 tubes AFS (for Santos CMI); dual exchangers 12″ x 20′ and 20″ x 20′ (for Canacol)
- Three-phase separator 72″ x 20′ ANSI 300 (for Conspecsime); well testing equipment TEA, FWKO, KOD, electrostatic treater (for Gran Tierra)
- Mechanical calculation reports for Ecopetrol: separators LV/LLV, pipe velocity, erosion rate, shell thickness, exchangers, pumps, vessel volume

WEATHERFORD INTERNATIONAL — TECHNICAL SERVICE AND PROJECT ENGINEERING

- Technical proposals for mechanical and process packages; 3D presentations for equipment and field layouts
- Mechanical layouts and pipe racks (4″ SCH XXS, Fig 602, 2.5 x 6 m, 9 t); reaction forces in an 8 MMSCFD relief line
- Anchor blocks and restraint bars (9 t) with Inventor frame analysis; welding plans for XXS connections at 1000 psi
- Pressure testing procedures for separation skids; conversion of skids into mobile crude loading stations
- Valve system upgrades for 10,000 psi separators; NACE construction procedure selection
- Flare system upgrade (18 m) to 16 MMSCFD with automatic ignition; flare radiation analyses for German sites near Hannover
- P&ID and mechanical layout standardisation across projects; flowback systems (12 tanks x 500 BBL)
- Dual sand filters and sand trap simulations (ANSYS); flowline simulation of 700 m for hydrate formation and KOD pressurisation
- Hydrate studies for relief valve activation (+15); temperature-loss profiles (+30); temperature, noise and radiation studies (+10)
- Orifice plate flow calculations; phase behaviour in choke valves
- Retention times in pressurised separators, gun barrels and tanks; erosion-rate prediction sheets

NEOCARBON — CLIMATE TECHNOLOGY

- Process development and scale-up of direct air capture (DAC) and CO_2 separation technology
- Equipment sizing, mass and energy balances, process simulation and technology evaluation

EDUCATION, CREDENTIALS AND LANGUAGES

- MSc Engineering Management – Berlin School of Business & Innovation / UNINETTUNO (2026), Sustainability and Energy Management, 90 ECTS
- MSc Engineering – Universidad Pontificia Bolivariana, Medellín (2023), GPA 1.6 German scale
- BSc Chemical Engineering – Universidad Pontificia Bolivariana (2020), thermodynamics research group
- Languages: Spanish (native), English (professional, IELTS Academic), German (B1)